Accessible Transportation

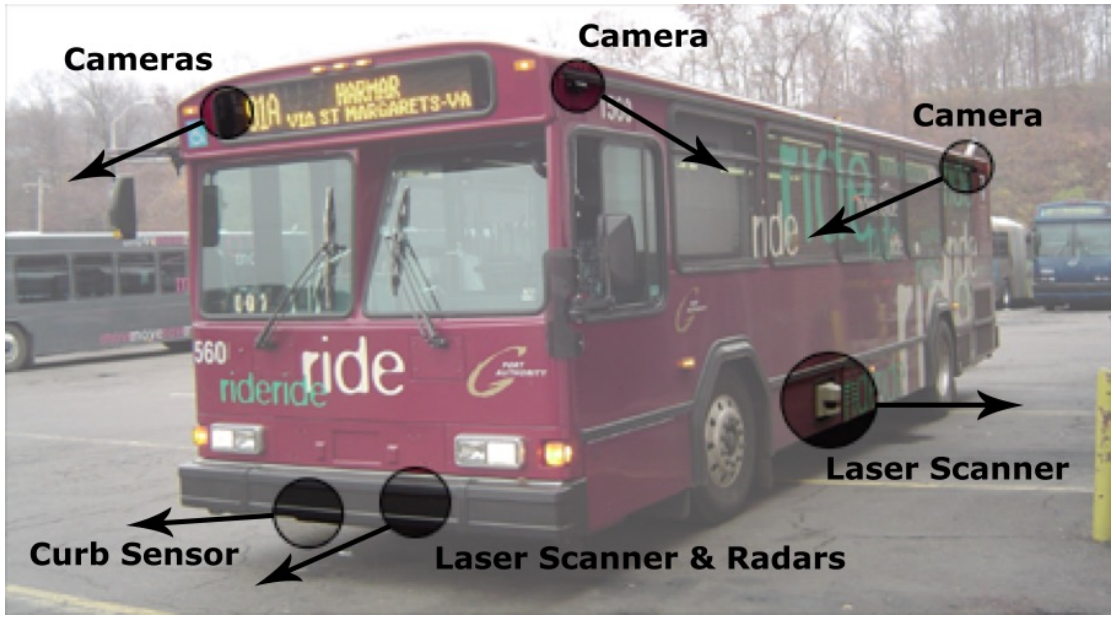
Aaron Steinfeld, *et al*

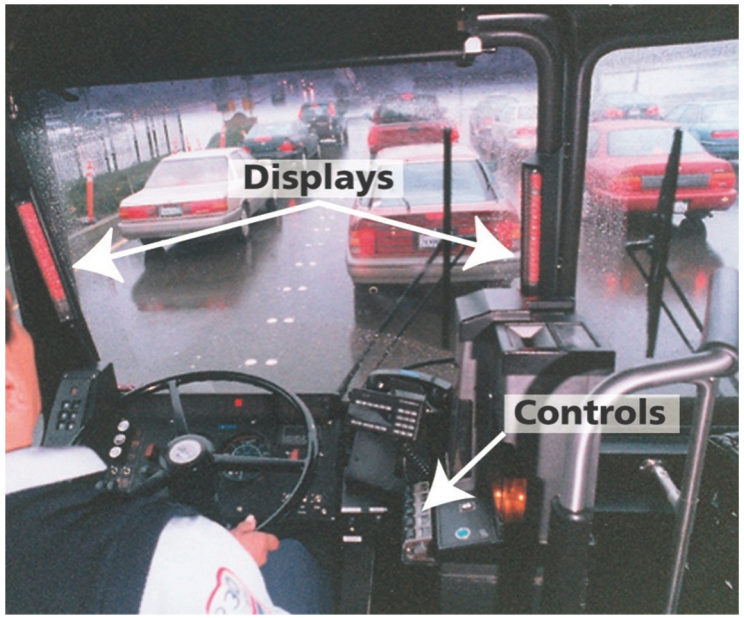
Transportation Bots & Disabilities (TBD) Lab

Robotics Institute, School of Computer Science

tbd.ri.cmu.edu

Early 2000s: Integrated Collision Warning System





Et al: TBD Lab and Collaborators

Many collaborators within CMU

- John Zimmerman
- Anthony Tomasic
- Elizabeth Carter
- Patrick Carrington
- Jeff Bigham
- Wenshan Wang
- RERC-APT
- Metro21, Traffic21, & Safety21
- Students, staff, and faculty

Many collaborators outside CMU

- University at Buffalo IDEA Center
- IBM Research
- University Washington
- Pittsburgh Regional Transit (PRT)
- VisAbility (aka Blind and Vision Rehabilitation Services of Pittsburgh)
- … and others

Universal Design

Transit, when done well, is a great example of UD

AI is also inherently compatible with UD (examples)

UD Principles

- Equitable Use
- Flexible Use
- Simple and Intuitive Use
- Perceptible Information
- Tolerance for Error

Goals of UD

- Awareness
- Understanding
- Social Integration
- Personalization



Interdisciplinary, User-Focused Innovation

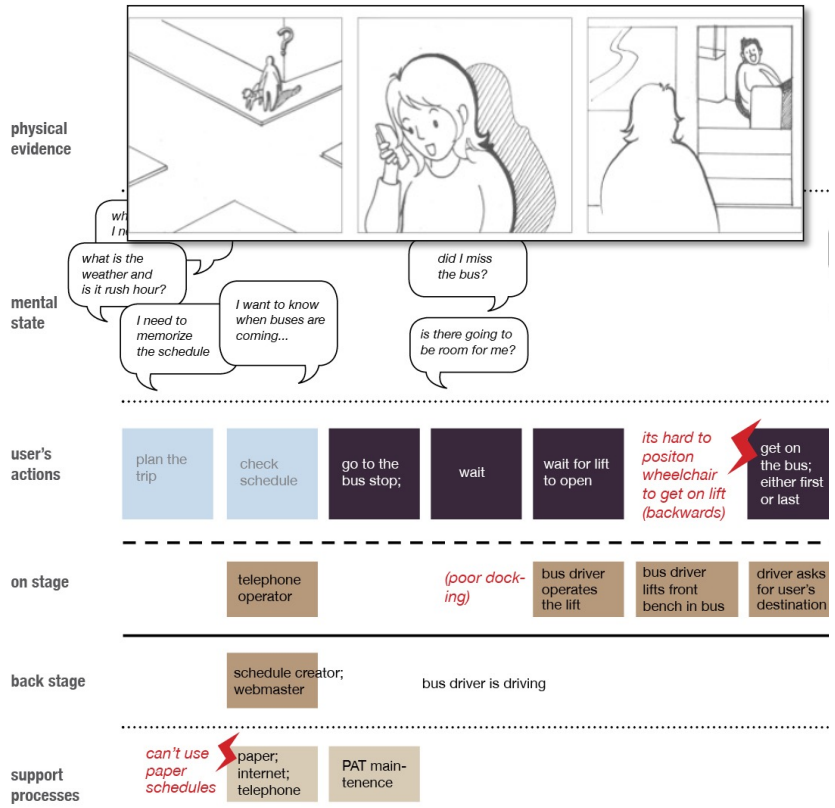
Understand the scenario

- Costs and ramifications for error
- Societal factors

AI should be "unremarkable" to users

Test, test, and test some more

- Real users and real-world



Tiramisu: Deployed, Machine Learning Testbed

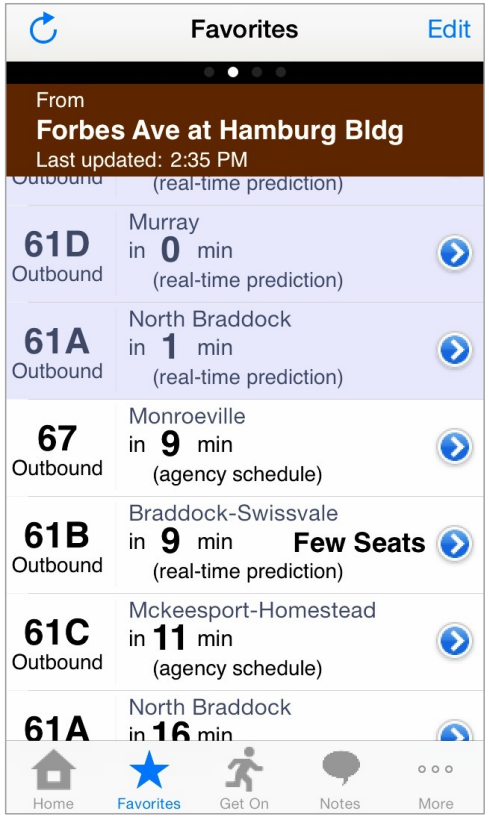
Version 1: crowdsourced information
- Real-time arrival, fullness

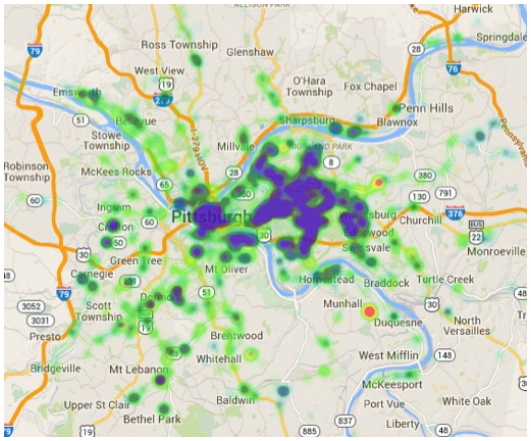
Version 2: added dialog and discussion

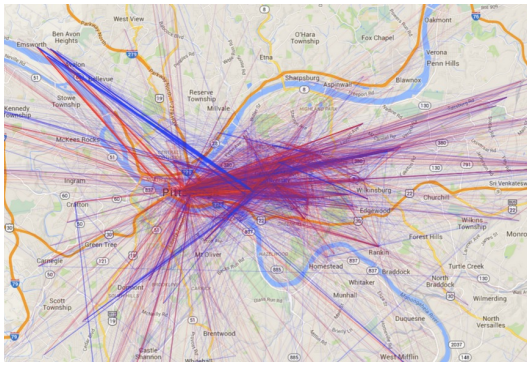
Version 3: personalized information

Fully deployed testbed 2011-2022
- Active in Pittsburgh and (some) NYC
- Millions of user-days

Analysis included both user behavior and post-hoc data mining
- Rider perceptions of fullness vs. APC
- Connection quality







Real-Time Crowdsource Performance

10 months of data
10,576 unique devices
296,283 interactive sessions

Users shared 28,257 location traces
Averaged 92 traces per day
Shared location traces for 10.54% of their sessions

Model	Abs. Error Mean (seconds)	Abs. Error S.D. (seconds)
Schedule	325	345
Historical	242	256
Real-time	123	151
Schedule rush hour	346	348
Historical rush hour	254	253
Real-time rush hour	122	151

119 seconds is the magic number

Performance:

- When there is data, good estimation accuracy
- Sparse coverage over whole PAAC system
- Better during rush hour; need a crowd for crowdsourcing
- During rush hour, users received real-time or historical model data for 23% of their information requests

Motivating User Contribution

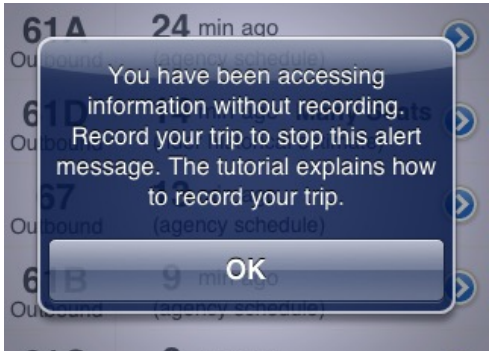
Users randomly assigned to conditions

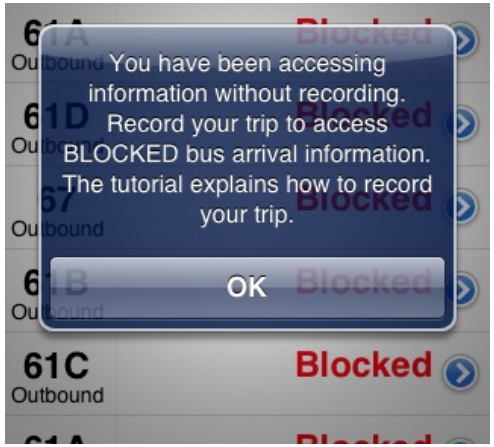
- Control
- Request
- Quid-pro-quo

8,447 participants over 10 months

Participants in:

- QPQ contributed more
- QPQ abandoned the app faster
- QPQ complained about in the app store
- Request similar to Control





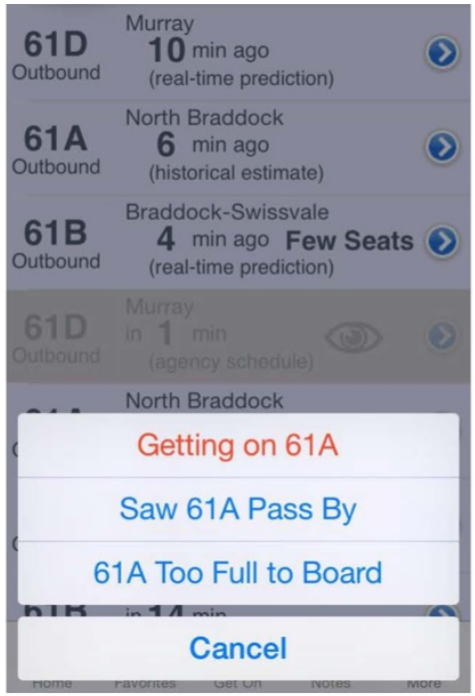
Lower the Bar for Crowdsourcing

Added spotting feature

- User requested feature
- “Getting on” for regular tracing
- “Pass By” for buses you don’t want
- “Too Full to Board” for buses you want that pass without letting you on

Led to improved app use and crowdsourcing

- Contributions increased coverage
- More app use
- Increase in regular tracing



Data Mining: Poor Connections

From a dataset with 60,466 trips

- Manually labelled 3,198 trip pairs
- Excluded round trips, bus turnarounds, and other issues
- 350 transfers from 145 devices

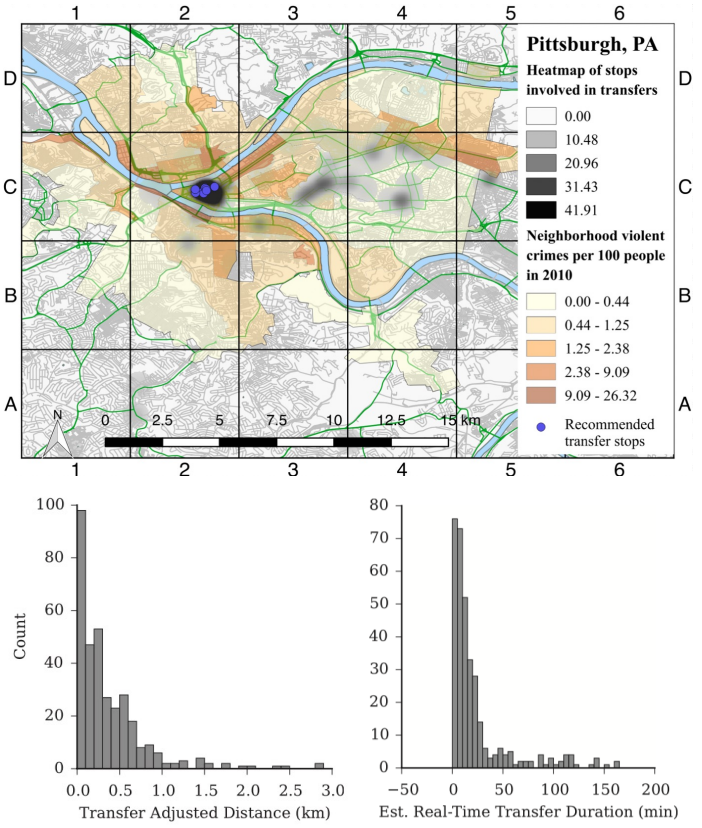
Long distances

- Inclement weather
- Missed connections

Infrastructure support

- Exposure to weather
- Exposure to crime

Provide visibility on poor transfers



Traut & Steinfeld, TR Part A, 2019

Carnegie Mellon University

Tiramisu v3: Predictive Route Filtering

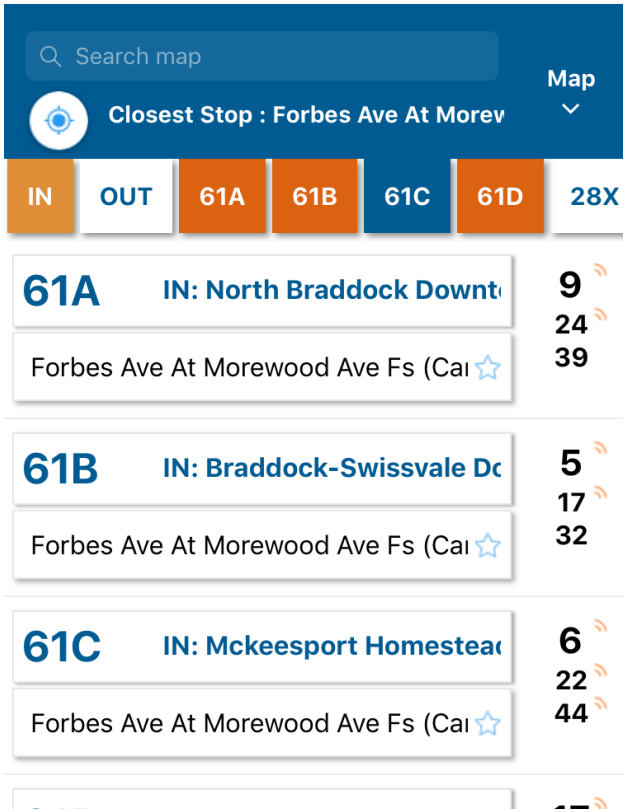
Machine learning built personalized model based on time & location

Filters selected by users (blue)

Pre-selected filters (orange)

Adaptive approach was successful (including screen reader users)

Users easily corrected system mistakes



Transparency on AI Behavior

Why a system made a recommendation

- "System suggestion"
- "Suggested based on past choices"

Vision and/or mobility-related disabilities

- Similar amount of time
- Preferred and rated as easier to use

"I chose the 61 bus with the shortest time, **the fact that it said I had previously used it,** and that it had available accessible seating." (mobility/motor-related disability)



Conversational Agents + Computer Vision

Mix of symbolic AI and machine learning

- Symbolic: Case-based reasoning, logic, etc
- ML: ChatGPT, GPT-3, etc

Speech interface for trip selection and guidance

- User-defined information (“downtown”)

Computer vision for nearby awareness

- Infrastructure-free location
- Identifying objects and reading signs



Carnegie Mellon University

"Experiment in Conversational Bus Stop Navigation for the Blind"

Carnegie Mellon University
Oscar J. Romero, Anthony Tomasic, John Zimmerman, Aaron Steinfeld
Copyright © 2024

Accessible Shared Autonomous Vehicles

Last mile applications

Semi-novel service issues

- People with disabilities
- Minors

Key issues

- Physical access to vehicles
- Assistance and awareness



Prior Work on Ramps and Full-Size Bus Layouts





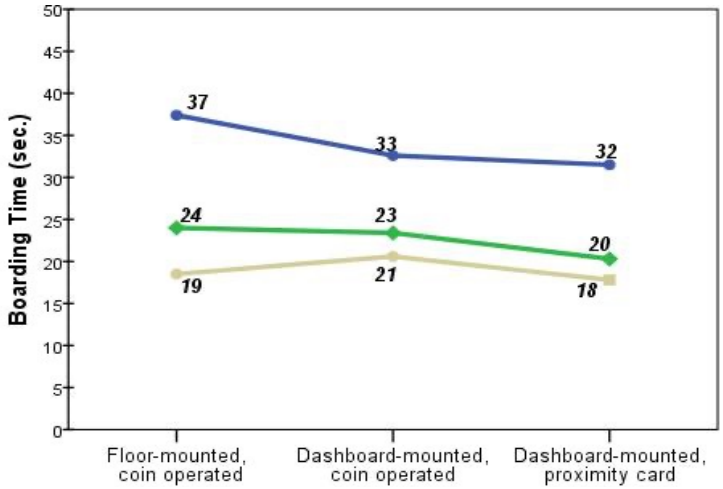
Fare Payment and Maneuvering Clearances



Floor-mounted fare machine



Dashboard-mounted fare machine



Floor-mounted fare-box design

- Space constraint for wheeled mobility users
- Easier to detect by cane users

Dashboard-mounted

- Proximity cards over coin
- Greater % of task completion

Swipe card was faster

Forward Facing Fare Machines



Extend the Reach of Transportation Workers

Telepresence into transit equipment

- Remote call and actuation
- Need to pinpoint customer location

Precise rendezvous is tricky

- Works in reverse too: fare machines, kiosks, AVs

Handoffs between static and mobile systems

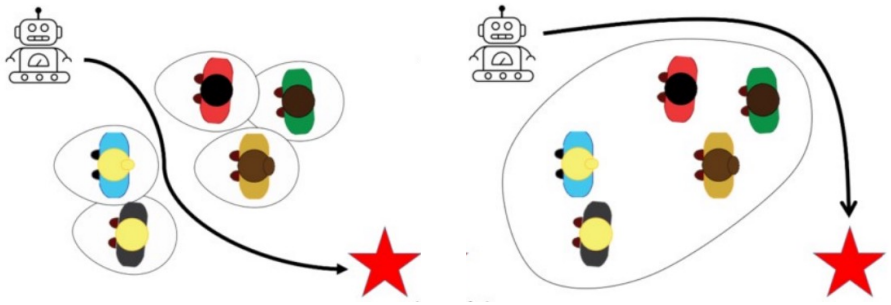


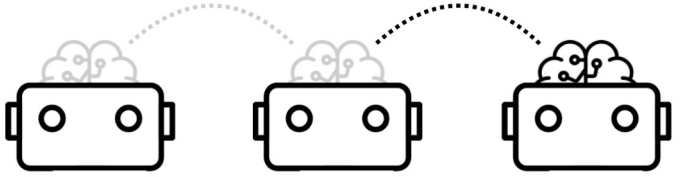


Examples from Korea

Many Robotics Challenges





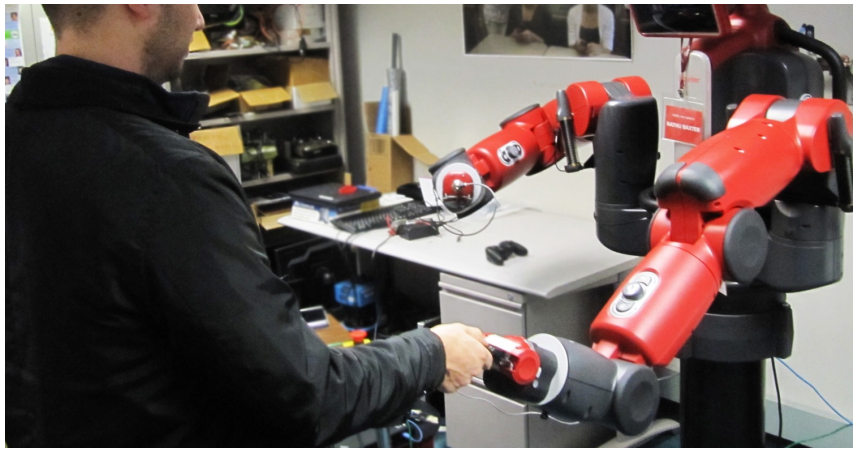


Local Robot Guides

Shared robots that live in a building

Inform and guide people to desired locations

Ensure safe and acceptable travel







Personal Guide Systems

User-owned systems to help with navigation

Computer vision, mapping, route planning, people detection

- Fine grained guidance and warnings
- Matching to building maps





Tan et al (left); CMU Cognitive Assistance Laboratory (right)

Accessible Delivery Robots

What people with disabilities need and want regarding deliveries

Appropriate behavior

- Understanding delivery locations
- Position in tight areas and near doors

Physical accessibility

Rendezvous

- Approach in larger spaces
- AVs at curb (similar to rideshare)



Accessible Delivery Robots

Workshop Study





Generating home delivery scenarios for semantic perception training and testing



Closing Thoughts

Partners, partners, partners

Stakeholders can make or break new systems

The value of AI and technology is directly linked to end user experience

Test, test, test

Benefits may come from unexpected places

Society's expectations for AI and robots are changing fast

The RERC-APT was funded by the National Institute on Disability, Independent Living, and Rehabilitation Research (NIDILRR grant numbers 90RE5011-01-00 and 90REGE0007; NIDRR grants H133E080019 and H133E130004). NIDILRR is a Center within the Administration for Community Living, Department of Health and Human Services.

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